

Formula sheet for Quiz 4

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more formulas than any one question needs, so deciding which applies is still your job.

A layer, forward

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}, \quad \mathbf{a} = \sigma(\mathbf{z}), \quad \mathbf{W} : (n_{\text{out}} \times n_{\text{in}}), \quad \mathbf{b} : (n_{\text{out}})$$

The chain rule

$$\frac{d}{dx}[f(g(h(x)))] = f'(g(h(x))) g'(h(x)) h'(x) \quad \frac{\partial \mathcal{L}}{\partial u} = \sum_k \frac{\partial \mathcal{L}}{\partial v_k} \frac{\partial v_k}{\partial u}$$

The second is the multivariable form, for a u that reaches the cost through several intermediate quantities $v_1, \dots, v_k$.

Local rules at a node

Let $g = \partial \mathcal{L} / \partial(\text{output of this node})$ be the gradient arriving from downstream.

$$\begin{aligned} c = a + b &\rightarrow a \text{ and } b \text{ each receive } g \\ c = a \times b &\rightarrow a \text{ receives } b g, \quad b \text{ receives } a g \\ c = \max(0, a) &\rightarrow a \text{ receives } g \text{ if } a > 0, \text{ else } 0 \\ c = f(a) &\rightarrow a \text{ receives } f'(a) g \end{aligned}$$

A layer, backward

Given $\partial \mathcal{L} / \partial \mathbf{z}$ handed down from the next layer:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}} \mathbf{x}^\top \quad \frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}} \quad \frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}}$$

Activations, and their derivatives

$$\begin{aligned} \sigma(z) &= \frac{1}{1 + e^{-z}} & \sigma'(z) &= \sigma(z)(1 - \sigma(z)) \\ \tanh(z) &= \frac{e^z - e^{-z}}{e^z + e^{-z}} & \tanh'(z) &= 1 - \tanh^2(z) \\ \text{ReLU}(z) &= \max(0, z) & \text{ReLU}'(z) &= 1 \text{ where } z > 0, 0 \text{ where } z < 0 \end{aligned}$$

Losses

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad \text{BCE} = -\frac{1}{n} \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

One example's BCE derivative: $\partial \ell / \partial \hat{p} = (\hat{p} - y) / [\hat{p}(1 - \hat{p})]$.

Gradient descent

$$\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} - \lambda \nabla \mathcal{L}(\mathbf{w}_{\text{old}}) \quad \text{iterations per epoch} = \left\lceil \frac{N}{B} \right\rceil$$

Notation

$\nabla_{\mathbf{x}} F$ is the gradient of F with respect to $\mathbf{x}$; θ is the full set of network parameters; ℓ is a single example's loss and $\mathcal{L}$ is the cost; J appears in place of $\mathcal{L}$ in some sources.